DRAFT BOX

Match Engine & Tactics Rulebook

How teams actually play against each other — every rating, formula, and formation matchup used by the match simulator, laid out exactly as implemented.

Draft Box V 1.1.4 — github.com/DeepInkGroup/draft-box

1. Overview

Every match in Draft Box — bot vs. bot, human vs. bot, or human vs. human — is decided by the same engine. It has three ingredients:

- **Player quality.** The overall rating of each of the 11 drafted (or bot-picked) players, weighted by how advanced or deep their exact slot sits on the pitch.
- **Formation shape.** A tactical fingerprint derived purely from the formation's slot layout — how attack-minded it is, how defensively solid it is, and how wide it stretches the pitch.
- **Chance.** Goals are sampled from a Poisson distribution around each team's "expected goals" number, so the better team wins more often — not every time.

Nothing here is hardcoded per formation or per team — every number below is derived from the same slot coordinates used to draw the pitch diagram in the app, so the numbers in this document are exactly what the live server computes.

2. The 16 Formations

Every formation is an explicit 11-slot layout (not just a GK/DF/MF/FW count) — this is what lets a player choose exactly which side of defense, midfield, or attack a drafted player fills.

Formation	GK	DF	MF	FW	Style
4-3-3	1	4	3	3	Attacking with width. Three forwards create constant threat.
4-4-2	1	4	4	2	Balanced and reliable. Two banks of four with a strike partnership.
4-2-3-1	1	4	5	1	Defensive solidity with a creative No.10 behind a lone striker.
4-5-1	1	4	5	1	Ultra-compact midfield. Absorb pressure and hit on the counter.
3-4-3	1	3	4	3	High-risk, high-reward. Bold width in attack, exposed at the back.
3-5-2	1	3	5	2	Wing-backs provide width while a five-man midfield dominates the center.
5-4-1	1	5	4	1	Fortress at the back. Sit deep, stay compact, strike on the break.
4-1-2-1-2	1	4	4	2	A narrow diamond in midfield builds through the center.
4-4-1-1	1	4	4	2	A second striker links midfield and attack just behind the target man.
5-3-2	1	5	3	2	Three at the back become five in defense. Cautious, hard to break down.
3-4-1-2	1	3	5	2	A free-roaming No.10 supports a strike duo behind a solid back three.
4-2-2-2	1	4	4	2	Two holding midfielders shield the back four while two playmakers link the strikers.
5-2-3	1	5	2	3	A back five with only two in the middle — firepower up top, engine room can get overrun.
5-3-1-1	1	5	3	2	Extreme defensive solidity, a target man and support striker linking play.
3-2-4-1	1	3	6	1	A back three with a double pivot, exploding into four attacking outlets.
4-2-4	1	4	2	4	Four forwards, two midfielders. Direct but leaves acres of space to exploit.

3. Formation Shape (the tactical fingerprint)

Three numbers are derived once per formation, straight from its slot list — no per-formation tuning:

```
defShape = SUM(every DF slot) + SUM(every MF slot with y >= 55, i.e. sitting deep)
atkShape = 1.5 × SUM(every FW slot) + SUM(every MF slot with y <= 35, i.e. pushed forward)
width = (max x) - (min x) among outfield slots
```

(y is pitch depth: 0 = the opponent's goal line, 100 = your own goal / GK. x is 0-100 left-to-right.) A formation with a back five and only one striker (5-4-1) has very high defShape and very low atkShape. A front four with two holding midfielders (4-2-4) is the exact opposite.

Formation	atkShape	defShape	width
4-3-3	4.5	4	70
4-4-2	3.0	4	70
4-2-3-1	4.5	6	70
4-5-1	1.5	5	76
3-4-3	4.5	3	72
3-5-2	3.0	4	80
5-4-1	1.5	5	80
4-1-2-1-2	4.0	5	70
4-4-1-1	3.0	4	70
5-3-2	3.0	5	80
3-4-1-2	4.0	3	72
4-2-2-2	5.0	6	70
5-2-3	4.5	5	80
5-3-1-1	3.0	6	80
3-2-4-1	5.5	5	76
4-2-4	6.0	4	70

4. Player Ratings

Every filled slot gets a continuous attack/defense weight from its own pitch depth (y):

```
attackWeight(slot) = 1 - y / 100
defenseWeight(slot) = y / 100
```

A striker (y~14) is weighted ~0.86 toward attack and only ~0.14 toward defense. A center back (y~78) is the reverse. A team's two headline numbers are then a weighted average of its 11 players' overall ratings:

```
TeamAttack = SUM(overall_i × attackWeight_i) / SUM(attackWeight_i) + starBonus
TeamDefense = SUM(overall_i × defenseWeight_i) / SUM(defenseWeight_i) + starBonus
```

Star Bonus

starBonus = min(3, numRealStarPlayersOnTheXI) × 1.5, added to both Attack and Defense. A squad with three or more of the handful of real, well-known stars seeded into the data gets a flat +4.5 rating lift, reflecting composure and quality that a raw overall average under-counts. A fourth star adds nothing further — the bonus caps out.

Captain Bonus (room-optional)

When a room has captains enabled, every member picks one of their own 11 players as captain once their draft is complete — the whole room must confirm a captain (if the option is on) before the World Cup can begin. That player's overall is boosted before the weighting above:

```
effectiveOverall(captain) = min(99, overall + 6)
```

Bots never have a captain — this is a reward for the deliberate choice a human squad-builder makes.

5. Chemistry

Beyond individual quality, how well an XI plays as a unit matters. Chemistry is three parameters, each 0-1, averaged into one score:

```
Linkage (L) = share of "adjacent" pitch partnerships sharing a real source team
Balance (B) = clamp( 1 - stdDev(overalls) / 20 , 0, 1 )
Leadership (S) = min(1, starPlayersOnXI / 3)
Chemistry = (L + B + S) / 3
multiplier = 0.94 + Chemistry × 0.12
```

The multiplier applies to both Attack and Defense, after the captain bonus. A 0.94-1.06 range means chemistry, like the formation edge, is a real but modest swing — it rewards a cohesive XI without letting it override raw quality.

5.1 Linkage — who's "adjacent"?

Every formation's slot layout defines a set of local partnerships: two slots within a short distance of each other on the pitch (a center back and the full back beside him, a central midfield trio, and so on — the same x/y coordinates used everywhere else in this document, no separate data). Linkage is the fraction of those partnerships where both players were drafted from the same real national team.

5.2 Worked example

An XI built entirely from Brazil's real squad (bots always draw their whole XI from one nation) versus a "franken-XI" of the same average overall (85) assembled from 11 different countries, both in a 4-3-3:

Squad	Linkage	Balance	Leadership	Chemistry	Multiplier
Real national squad	1.00	0.89	1.00	0.96	×1.056
11-country all-star XI	0.00	1.00	1.00	0.67	×1.020

Simulated 3,000 times at equal overall (85 each): the real national squad won 54% of the time to the all-star XI's 22% (rest drawn). This is a deliberate trade-off, not a bug: bot-controlled real countries are guaranteed perfect Linkage, so a human squad built purely by chasing the highest overalls across every country in the draft pool will often be trading chemistry away for star power — both are viable strategies.

6. Match Simulation

6.1 Formation Edge

How much team X's attacking shape overwhelms team Y's defensive shape, plus a bonus for X being meaningfully wider than Y:

```
edge(X->Y) = clamp( (atkShape_X - defShape_Y) × 0.4 + (width_X - width_Y) × 0.06 , -6, +6
)
```

Worked example (pure width, everything else equal). 4-4-2 and 3-5-2 have identical atkShape (3.0) and defShape (4) — the only difference is width (70 vs. 80). Yet:

```
edge(3-5-2 attacking -> 4-4-2 defending) = +0.20
edge(4-4-2 attacking -> 3-5-2 defending) = -1.00
```

3-5-2's extra width alone is worth a full 1.2-point swing in its favor — stretching a narrower back four creates gaps a flat 4-4-2 doesn't have to worry about defending.

6.2 Expected Goals

```
xG(X vs Y) = clamp( 1.35 + (Attack_X - Defense_Y) / 18 + edge(X->Y) / 10 , 0.15, 4.5 )
```

Both teams' xG are computed independently and simultaneously (X attacking into Y's defense, and Y attacking into X's defense) — note the formation edge divides by 10 while the base 1.35 and player-quality term dominate the number. This is deliberate: the tactical matchup nudges the scoreline by roughly half a goal at most; player quality is still the primary driver of most results.

6.3 Goals

The actual scoreline is sampled from a Poisson distribution using each team's own xG as the mean — the standard way to turn an "expected goals" number into a real, whole-number, occasionally-surprising result.

6.4 Knockout draws -> penalties

```
composure = Attack + Defense
P(X wins shootout) = clamp( 0.5 + (composure_X - composure_Y) / 400 , 0.35, 0.65 )
```

Penalties are mostly a coin flip by design (clamped to a 35-65% range) — the better side gets a real but modest edge, never a lock.

6.5 Red Card Morale Hit

A player sent off rattles their own team for a short spell afterward, regardless of that individual player's own quality. Since the engine decides a match in one shot rather than minute-by-minute, the "short spell" is expressed as a fraction of the full 90 minutes and converted into an average-over-the-match Attack + Defense reduction, applied before the Expected Goals step (6.2):

```
P(red card this match) = 7%, assigned to a random outfield player on a random side
affectedMinutes = 15, or 15 × 1.8 = 27 if the sent-off player is the captain
moraleImpact = 0.20 × (affectedMinutes / 90)
Attack, Defense ×= (1 - moraleImpact)
```

That works out to roughly a 3.3% temporary rating cut for an ordinary player sent off, or about 6% if the captain is the one dismissed — small individually, but it measurably lowers a red-carded side's average goals scored over many matches. Bots never have a captain (Section 4), so this only doubles up for human squads that chose one.

7. Formation Matchups

Holding player quality equal, here is every formation's best and worst matchups by pure formation edge (positive = advantage attacking into that opponent's defense). Remember: this is a nudge on top of player quality, not a guarantee — a Brazil XI in a "weak" matchup will still usually beat a Jordan XI in a "strong" one.

Formation	Strong against	Struggles against
4-3-3	3-4-3 (+0.5), 3-4-1-2 (+0.5), 4-4-2 (+0.2)	5-3-1-1 (-1.2), 5-2-3 (-0.8), 5-3-2 (-0.8)
4-4-2	3-4-3 (-0.1), 3-4-1-2 (-0.1), 4-3-3 (-0.4)	5-3-1-1 (-1.8), 5-2-3 (-1.4), 5-3-2 (-1.4)
4-2-3-1	3-4-3 (+0.5), 3-4-1-2 (+0.5), 4-3-3 (+0.2)	5-3-1-1 (-1.2), 5-2-3 (-0.8), 5-3-2 (-0.8)
4-5-1	3-4-3 (-0.4), 3-4-1-2 (-0.4), 4-3-3 (-0.6)	5-3-1-1 (-2.0), 5-2-3 (-1.6), 5-3-2 (-1.6)
3-4-3	3-4-1-2 (+0.6), 4-3-3 (+0.3), 4-4-2 (+0.3)	5-3-1-1 (-1.1), 5-2-3 (-0.7), 5-3-2 (-0.7)
3-5-2	3-4-3 (+0.5), 3-4-1-2 (+0.5), 4-3-3 (+0.2)	5-3-1-1 (-1.2), 5-2-3 (-0.8), 5-3-2 (-0.8)
5-4-1	3-4-3 (-0.1), 3-4-1-2 (-0.1), 4-3-3 (-0.4)	5-3-1-1 (-1.8), 5-2-3 (-1.4), 5-3-2 (-1.4)
4-1-2-1-2	3-4-3 (+0.3), 3-4-1-2 (+0.3), 4-3-3 (+0.0)	5-3-1-1 (-1.4), 5-2-3 (-1.0), 5-3-2 (-1.0)
4-4-1-1	3-4-3 (-0.1), 3-4-1-2 (-0.1), 4-3-3 (-0.4)	5-3-1-1 (-1.8), 5-2-3 (-1.4), 5-3-2 (-1.4)
5-3-2	3-4-3 (+0.5), 3-4-1-2 (+0.5), 4-3-3 (+0.2)	5-3-1-1 (-1.2), 5-2-3 (-0.8), 5-4-1 (-0.8)
3-4-1-2	3-4-3 (+0.4), 4-3-3 (+0.1), 4-4-2 (+0.1)	5-3-1-1 (-1.3), 5-2-3 (-0.9), 5-3-2 (-0.9)
4-2-2-2	3-4-3 (+0.7), 3-4-1-2 (+0.7), 4-3-3 (+0.4)	5-3-1-1 (-1.0), 5-2-3 (-0.6), 5-3-2 (-0.6)
5-2-3	3-4-3 (+1.1), 3-4-1-2 (+1.1), 4-3-3 (+0.8)	5-3-1-1 (-0.6), 5-3-2 (-0.2), 5-4-1 (-0.2)
5-3-1-1	3-4-3 (+0.5), 3-4-1-2 (+0.5), 4-3-3 (+0.2)	5-2-3 (-0.8), 5-3-2 (-0.8), 5-4-1 (-0.8)
3-2-4-1	3-4-3 (+1.2), 3-4-1-2 (+1.2), 4-3-3 (+1.0)	5-3-1-1 (-0.4), 5-2-3 (-0.0), 5-3-2 (-0.0)
4-2-4	3-4-3 (+1.1), 3-4-1-2 (+1.1), 4-3-3 (+0.8)	5-3-1-1 (-0.6), 5-2-3 (-0.2), 5-3-2 (-0.2)

Reading this: very defensive back-five shapes (5-3-1-1, 5-2-3, 5-3-2) are hard for anyone to break down, so they show up as the toughest matchup across the board. Wide, attack-heavy shapes (4-2-4, 3-2-4-1, 5-2-3) punish narrow or thin-on-defense formations (3-4-3, 3-4-1-2) the hardest. Bots are assigned a random formation for every tournament, so these matchups play out differently every run.

8. Tournament Format & Room Options

8.1 Full Tournament (default)

- 48 teams → 12 groups of 4, drawn at random.
- Each group plays a round robin (3 matchdays, 6 matches).
- Top 2 of each group (24) + the 8 best third-placed teams advance to the Round of 32 — the real 2026 FIFA World Cup format.
- Round of 32 → Round of 16 → Quarter-Finals → Semi-Finals → Final, single elimination.
- Any knockout draw goes to penalties (formula in Section 6.4).

8.2 Tournament Length (room-optional)

The room creator picks one of three lengths before the draft starts. Every human drafter is guaranteed a slot in whichever bracket size is chosen; bot teams are randomly trimmed down from the full 48 to fill the rest.

Length	Starting teams	What happens
Off (default)	48	Full group stage → Round of 32 → ... → Final
Blitz	32	Skips the group stage entirely, starts directly at the Round of 32
Top 8 (1/4)	8	Skips straight to the Quarter-Finals with only 8 teams

8.3 Restrict Draft Teams (room-optional)

The room creator can optionally limit which nations are eligible to be revealed during the draft to a hand-picked list of at least 4 teams (leave it empty for the default: all 48 teams eligible). This only affects which teams appear during drafting — the World Cup simulation itself still fields all of the tournament's bot and human slots as normal. Every nation reveals independently to each drafter, so if two members of a room are both shown the same country at the same time, they each pick from that team's remaining available players on their own — one member's pick never blocks or disrupts the other's reveal.

9. Live Match Reports

Every simulated match generates an event timeline — goals with scorer and (usually) an assist, plus yellow/red cards — using the same slot-weighting logic as everywhere else in this document: scorers are drawn from a team's XI weighted toward advanced (low-y) slots and higher overall, assists weighted toward midfield depth. Purely narrative — events don't feed back into the score, which is already decided by Section 6.

Results arrive one stage at a time: nobody is forced to watch every match in a round at once, and one person clicking through doesn't drag anyone else's screen forward. Each human's own match plays out on a live ticking 90-minute clock; every other match in that round is available as a static final report via a match picker. A pre-tournament screen also shows a championship-odds estimate and a predicted top scorer/assist for the viewer's own squad, derived from the same rating and slot-weighting formulas above.

9.1 Team Performance Stats

Every report — live or static — also shows possession, pass count, pass accuracy, and xG for both sides, derived straight from the same post-chemistry, post-red-card Attack/Defense numbers used for the Expected Goals step, not simulated independently:

```
qualityDiff = (Attack_A + Defense_A) - (Attack_B + Defense_B)
possession_A = clamp( 50 + qualityDiff × 0.6 , 32, 68 ); possession_B = 100 - possession_A
```

```
passAccuracy_X = clamp( 68 + (avgRating_X - 75) × 0.7 , 55, 94 ), avgRating =
(Attack+Defense)/2
totalPasses = 780 ± up to 60 (match tempo, random)
passes_X = round( totalPasses × possession_X / 100 )
```

Narrative, like the event feed — a lopsided possession or pass-accuracy split reflects the same quality gap that produced the scoreline, it doesn't independently sway it.

Appendix: Data Disclaimer

The 48-team list matches the real, official field for the 2026 FIFA World Cup, and **every single one of the 1,248 players in the game is a real member of that nation's actual 26-man 2026 World Cup squad** (sourced from Wikipedia's squad lists, themselves drawn from FIFA's official squad announcements). No fictional or generated players exist anywhere in the dataset — a team with fewer real players in a given position group than a demanding formation calls for simply won't be revealed to a drafter who still needs that position, and a bot fielding that formation with that nation falls back to reusing its weakest already-selected player rather than inventing one.

Overall ratings are fictional gameplay values, not an official or licensed rating dataset. See RULES.md Section 9 for the full breakdown.